SIDE QUEST HOTLINE · VAPI HACKATHON

CallLens

Voice agents fail silently. CallLens shows which turn broke, why, and fixes it live.

Karthik Raja Anandan · Vapi + Deepgram + Exa + TypeSafe Jev

Why this is necessary

Latency

#1 complaint in Vapi's forums: real latency up to 4× the estimate, 3–5 s pauses.

Accents

Speech-to-text slips on heavy accents and noise. Agents then loop and callers quit.

Blind spots

Users ask for latency breakdowns and decision traces. Logs say "customer ended call."

A failed call has no error. It just ends. You need to see inside it.

What I built tonight

THE AGENT

Side Quest Hotline

Call it, say your vibe, area and time. Exa searches live, Jev picks one real thing to do right now.

THE LENS

CallLens

Every call, every turn: latency by stage, what broke, Jev's verdict, and a one-click live fix.

Caller → **Deepgram** ears → **Vapi** orchestrates → **Exa** search → **Jev** decides → **Deepgram** voice

Architecture



CallLens loop

Vapi call API → per-turn latency + transcript → Jev judges every turn, the call, the language → dashboard

Self-heal

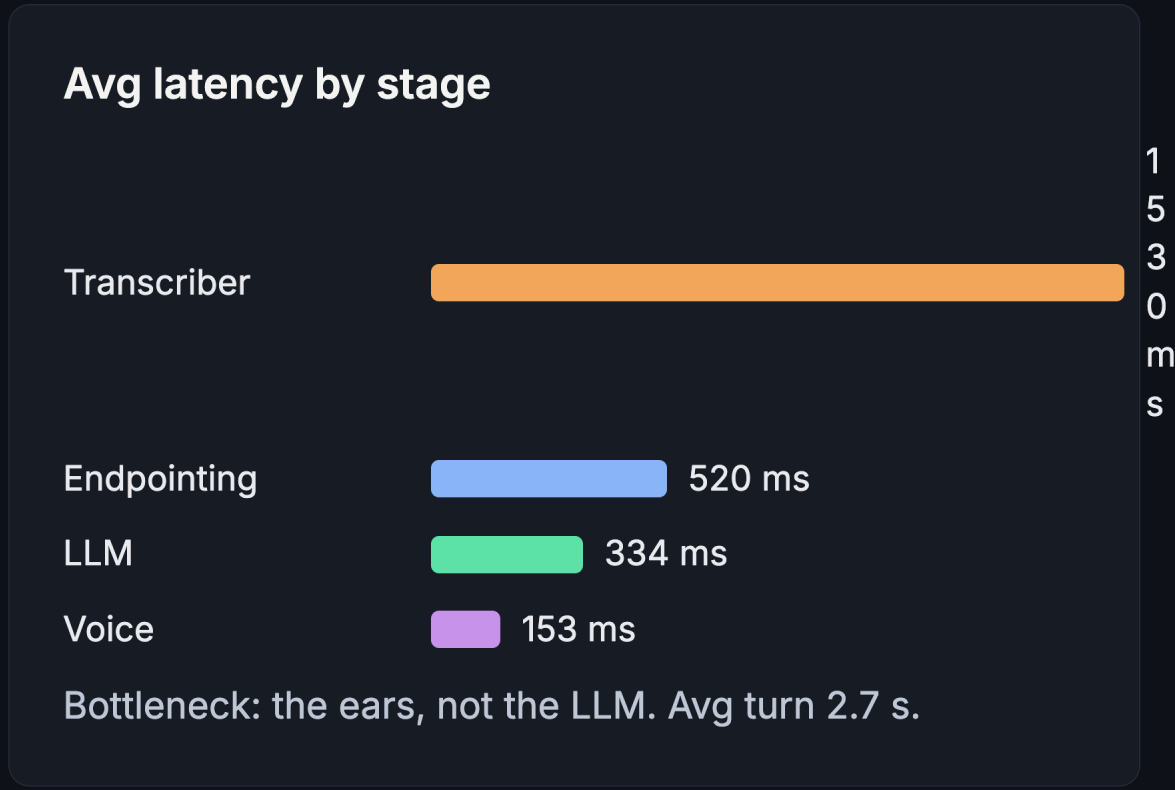
Findings → patch → Vapi PATCH /assistant (keyterms, endpointing, language) → next call verifies

One real call, from this room

I SAID “EMBARCADERO.” DEEPGRAM HEARD:

Aero.
Arrow.
Amir.

The agent asked my location three times. **10 of 22 turns** broke; worst gap **6.4 s**.



Jev: the fast judge

6 / 6

searches where Jev changed the pick vs Exa's #1

~110 ms

per decision: fast enough to run live, mid-call

5 jobs

pick, turn issue, call verdict, goal met, language

"Quiet coffee and books": without Jev → a reading party. With Jev → **Silent Reading Club (42% vs 17%)**. Jev's reasoning is its probability over every option.

I asked Vapi Composer too

Same call	Vapi Composer	CallLens
Root cause	Prompt: re-asks location	Transcriber misheard “Embarcadero”
Latency	Not mentioned	Transcriber 1530 ms = bottleneck
Provider fix	“No change needed”	Deepgram keyterm + 0.4 s endpointing
Speed	~45 s LLM agent	~120 ms per Jev judgment
Scope	One thread per question	Every call, trends, auto-fix

Composer is right about the prompt. CallLens sees the speech layer it can’t. **Best together.**

New features vs what exists

Feature	Today	CallLens upgrade
Latency per stage	Raw numbers per call	Names the bottleneck across calls
Turn diagnosis	Read logs yourself	Each turn labeled + fix
Call verdict	LLM analysis, seconds	Jev quality + goal met, ~120 ms
Language	Set by hand	Jev detects; heal sets multi
Tool decisions	LLM or first result	Jev Choice with probabilities
Fixing	Edit config manually	One-click live patch, verified

Self-heal, tested on a repeat call

Same caller, same script	Before heal	After heal
“Embarcadero” heard as	Aero · Arrow · Amir	Embarcadero, first try
Agent re-asked location	3 times	1 time
Turns to finish	22	17
Avg turn latency	2731 ms	2947 ms (not fixed)
Transcriber	1530 ms	1551 ms (next fix)

Patch applied live via Vapi API: Deepgram keyterm “Embarcadero” + shorter endpointing wait. **Mishearing fixed; latency is still transcriber-bound, and CallLens shows it.**

Vapi tells you what happened. CallLens tells you why, and fixes it.

Next: live mid-call repair (“Did you say Embarcadero?”), regression replay via Simulations, CallLens as an MCP server for your coding agent.

Call it: +1 (541) 777-8465 · I build evals and latency tooling for production agents. Let’s talk.